James Olaitan

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Education

Minerva University — San Francisco, CA
Bachelor of Science in Computer Science, GPA: 3.6/4.0 — *Sep 2023 - May 2027*
Coursework: Data Structures & Algorithms, Probability & Statistics, Statistical Modeling, Linear Algebra

Technical Skills

Languages: Python, C/C++, Rust, Go, JavaScript, SQL, R, Bash
Frameworks: Flask, React, FastAPI, PyTorch, SQLAlchemy, NumPy, Pandas, Matplotlib, Streamlit
Tools: Docker, Kubernetes, Git, CMake, Terraform, GitHub Actions, Grafana, Prometheus, GoogleTest
Cloud & Databases: AWS, Azure, Linux, PostgreSQL, MySQL, DuckDB
Certifications: Microsoft Azure Fundamentals, IBM Data Science Professional

Experience

Meta — Jun 2026 - Sep 2026
Production Engineer, SRE (MLH Fellow) | Linux, Docker, Flask, Prometheus, Grafana, CI/CD — *Remote*

- Built a Flask web service with MySQL on a Linux VPS, wrote Bash scripts for provisioning and deployment automation, and shipped through a GitHub Actions CI/CD pipeline for reproducible production releases.
- Configured NGINX as a reverse proxy for routing and SSL termination; instrumented the stack with Prometheus and Grafana for metrics and alerting; diagnosed live failures with grep, ps, and journald.

Google Summer of Code, Red Hat (7.4% acceptance) — May 2026 - Aug 2026
Open Source Contributor, Konflux (Red Hat) | Tekton, Go, buildah, Bash — *Remote*

- Authored Konflux's ADR-0069, a reproducible-build design where two independent builds of the same source must produce matching digests, making tampered container images impossible to ship undetected.
- Implementing ADR-0069 in Konflux's upstream build-definitions: 3 opt-in reproducibility flags wired across 6 docker-build* pipeline variants; verified end-to-end by two independent rebuilds of the same commit producing byte-identical digests.

UL Solutions — Apr 2024 - Aug 2024; May 2025 - Jul 2025
Engineering Intern | Linux, Bash, RF/EMC Testing — *Fremont, CA*

- Debugged firmware flashing pipelines across 30+ IoT variants in 3 FCC/ETSI-certified product lines, tracing calibration drift and script failures in Linux-based RF/EMC test environments to increase throughput by 20%.
- Flagged an emissions violation that led a client to scrap a planned product feature (preventing a federal compliance issue) while processing 500+ RF/EMC test results into CSV/DAT formats.

Projects

IAM Privilege Escalation Detector | *Go, AWS IAM, OPA/Rego, Terraform, Docker, GitHub Actions, Python*

- Detected multi-hop AWS IAM privilege escalation paths that per-policy static analyzers miss, by modeling IAM as a directed graph with 9 edge types and running BFS over offline snapshots without AWS calls.
- Discovered that security-tool benchmarks conflate two detection models, producing misleading accuracy numbers; built a recall benchmark across 3 tools and 15 scenarios with Wilson-score CIs, reproducible from a fresh clone without AWS credentials.
- Prevented AWS customer-identifier leaks (account IDs, ARNs) from shipping by adding a commit-time audit script that also enforces layer-dependency rules across the 205-test suite.

Systemd Prometheus Exporter | *Rust, Prometheus, systemd, Linux*

- Built a Rust daemon that polls systemd services using systemctl and exposes real-time service health as Prometheus metrics (service_up gauge, service_checks_total counter) over HTTP, enabling Grafana to alert on service failures.
- Implemented TOML-driven configuration, logfmt-structured logging on state changes, and graceful SIGTERM/SIGINT shutdown with no async runtime (4 crates), deployable as a systemd unit on any Linux host.

Market Anomaly Detection | *Python, PyTorch, DuckDB, Streamlit, Docker, GitHub Actions*

- Detected correlation breakdowns at a 0.04% base rate in S&P 500 sector data with 75% precision and 0.9998 ROC-AUC, by training an 18K-parameter PyTorch LSTM on 14 years of history.
- Prevented lookahead bias from silently inflating metrics by enforcing chronological train/val/test splits across a DuckDB pipeline of 5 tables and 8 rolling/z-score features, with 25 tests covering schema and feature integrity.

Organizations

ColorStack, Brilliant Black Minds